

Cecilia A. Callejas Pastor

AI Scientist in Healthcare

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RESEARCH INTERESTS

- Artificial Intelligence and Deep Learning for Medical Applications
- Computer Vision and Image Recognition Systems
- Biomedical Signal Processing and Machine Learning
- Mobile Health Applications and Wearable Devices
- Clinical Decision Support Systems

EXPERIENCE

Postdoctoral Researcher

Tokyo University, Department of Bioengineering

Tokyo, Japan

June 2026 - Present

- Conducting data science and AI research on healthy longevity, sleep epidemiology, and metabolic syndrome prevention, analyzing biodata, wearable-device data, and digital lifestyle data to identify health-risk patterns and support personalized feedback.

Research Professor

Chungnam National University Hospital, Biomedical Research Institute

Daejeon, South Korea

Dec 2025 - May 2026

- Leading research on LLM-driven tools for CKD and AKI management, developing models that interpret clinical text and structured data to flag high-risk patients, support individualized treatment decisions, and reduce progression to kidney failure.

Postdoctoral Researcher

Seoul National University, Department of Radiology

Seoul, South Korea

Sept 2024 - Nov 2025

- Leading AI-driven technology development with deep learning and computer vision algorithms, implementing recognition models and developing AI solutions using Python and PyTorch for real-time applications and automated systems.

Postdoctoral Researcher

Chungnam National University, Integrated Research Institute

Daejeon, South Korea

Sept 2024 - Mar 2025

- Developing hardware and firmware solutions for AI-integrated medical devices, focusing on embedded systems implementation with Java and Flutter for real-time clinical monitoring and diagnostic equipment.

Senior Researcher Intern

MedInTech Inc.

Seoul, South Korea

July 2024 - Aug 2024

- Developed web server for GPU monitoring, MedInTech Gateway software for endoscopic video capture, and engineered MD-SC-300 AI cancer classification system for medical devices.

Head of Biomedicine and Chemical Department

IDETECA S.R.L.

La Paz, Bolivia

Sept 2017 - June 2018

- Led software and hardware evaluation, testing, and troubleshooting for biomedical devices and sensor technologies.

Technical Service Department Engineer

IMEXCOMED S.R.L.

La Paz, Bolivia

May 2016 - July 2017

- Performed maintenance of endoscopic and surgical devices with software support and installation services in hospital environments.

EDUCATION

Ph.D. in Biomedical Engineering
Chungnam National University

Daejeon, South Korea
Sept 2021 – Aug 2024

Overall GPA: 99.2/100 - Global Korean Scholarship Program Scholar (Full Scholarship)

MSc in Biomedical Engineering
Chungnam National University

Daejeon, South Korea
Sept 2019 – Aug 2021

Overall GPA: 97.8/100 - Global Korean Scholarship Program Scholar (Full Scholarship)

BSc in Biomedical Engineering
Bolivian Catholic University

La Paz, Bolivia
Feb 2012 – June 2016

Overall GPA: 88.5/100 - Scholarship awarded for highest GPA in multiple semesters

Bolivian Catholic University
Diploma in Professor Training for Higher Education

La Paz, Bolivia
Feb 2016 – Aug 2016

TEACHING

Teaching Assistant
Chungnam National University

Daejeon, South Korea
Fall 2022

- Assisted with Basic Computer Programming and Biomedical Instrumentation courses including grading exams, conducting lab sessions, and providing student support.

Teaching Assistant
Chungnam National University

Daejeon, South Korea
Spring 2022

- Assisted with Biomedical Sensors course including exam grading, laboratory supervision, and student mentoring.

Teaching Assistant
Chungnam National University

Daejeon, South Korea
Fall 2020

- Assisted with Basic Computer Programming course including grading assignments and exams, tutoring students, and supporting laboratory activities.

Professor
Bolivian Catholic University

La Paz, Bolivia
Mar 2018 – Aug 2018

- Taught Electromagnetism and Calculus 1 courses.

SKILLS

Programming Languages: Python, MATLAB, Java, Dart/Flutter, C/C++

Software Development: Android Studio, REST API, Postman, Jira Cloud, Streamlit, FastAPI

Hardware Design: SolidWorks, OrCAD, PADS, Segger Embedded Studio, STM32 CubeMX

Languages: Spanish (Native), English (TOEIC 970), Korean (TOPIK 5), Italian (Basic)

Soft Skills: Hard-working, Attention to detail, Leader, Critical thinking, Responsibility, Collaboration

AWARDS AND ACHIEVEMENTS

Best Poster Award – Korean Balance Society	2025
Best Poster Award – Korean Innovative Medical Technology Society	2024
Best Oration Award – Korean Balance Society	2023
Best Poster Award – International Biomedical Engineering Conference	2021
Best Research Award – Chungnam National University	2021
Third Place, AI & BigData Hackathon – Daewoong Foundation	2021

PUBLICATIONS

Journal Articles

1. S Sabir, **CA Callejas Pastor**, Y Ku, C Oh. "Development of Personalized Integrated Hemodynamic Management Guidelines Using Multimodal Non-Invasive Monitoring for Postoperative Delirium" (First Co-Author). *British Journal of Anaesthesia* (Under Review).
2. C Kwon, D Shim, JS Joo, Y Go, **CA Callejas Pastor**, Y Ku, MW Suh. "Therapeutic efficacy of canalith repositioning procedure using by inertial measurement unit sensor-based guiding system for idiopathic benign paroxysmal positional vertigo" *Journal of Vascular Research* (Under Review).
3. H Ryu, **CA Callejas Pastor**, JS Joo, Y Ku, MW Suh. "An Advanced Rule-Based Mobile Classifier for the Automated Diagnosis of Vestibular Disorders" (First Co-Author). *Scientific Reports* (2026).
4. **CA Callejas Pastor**, Y Ku, SH Jeong, E Kwon. "Autonomic Nervous System Responses to Virtual Reality Exposure: Heart Rate Variability Dynamics During Simulated Roller Coaster Experience" *Scientific Reports* (2026).
5. **CA Callejas Pastor**, H Ryu, JS Joo, Y Ku, MW Suh. "Clinical Decision Support for Vestibular Diagnosis: Large-scale Machine Learning with Lived Experience Coaching" *NPJ Digital Medicine* (2025).
6. **CA Callejas Pastor**, C Oh, B Hong, Y Ku. "Machine learning-based cardiac index estimation using photoplethysmography in off-pump coronary artery bypass surgery" *Journal of Clinical Medicine* 13, 7145 (2024).
7. **CA Callejas Pastor**, C Kwon, JS Joo, HC Kim, Y Ku, MW Suh. "Feasibility of an inertial measurement unit sensor-based guiding system for the treatment of benign paroxysmal positional vertigo" *Scientific Reports* 13, 3169 (2023).
8. **CA Callejas Pastor**, IY Jung, S Seo, SB Kwon, Y Ku, J Choi. "Two-dimensional image-based screening tool for infants with positional cranial deformities: A machine learning approach" *Diagnostics* 10, 495 (2020).

Conference Presentations

1. I Karim, C Kwon, **CA Callejas Pastor**, Y Ku. "Non-invasive electro-acoustic stimulation system for tinnitus treatment" *IEEE EMBC 2025*, Copenhagen, Denmark.
2. I Karim, C Kwon, **CA Callejas Pastor**, Y Ku. "Design of a non-invasive bimodal auditory nerve stimulation circuit for tinnitus treatment" *International Biomedical Engineering Conference 2024 (IBEC)*, Seoul, South Korea.
3. C Kwon, E Kim, S Kim, **CA Callejas Pastor**, J Joo, MW Suh, Y Ku. "A study on feasibility of Benign Paroxysmal Positional Vertigo Treatment Assistive Devices and Quantitative analysis of the treatment risk factor" *International Biomedical Engineering Conference 2024 (IBEC)*, Wonju, South Korea.
4. **CA Callejas Pastor**, J Joo, H Ryu, MW Suh, Y Ku. "Development of a screening app for vestibular disorders: A promising tool for diagnosis" *IEEE EMBC 2023*, Sydney, Australia.
5. **CA Callejas Pastor**, H Ryu, J Joo, MW Suh, Y Ku. "A Rule-Based Mobile Application for the Screening of Vestibular Disorders: A Pilot Study" *Korean Equilibrium Society Conference 2023*, Seoul, South Korea.
6. **CA Callejas Pastor**, MW Suh, Y Ku. "Machine learning-based automated classification of vestibular disorders" *International Biomedical Engineering Symposium 2022 (IBES)*, Incheon, South Korea.
7. **CA Callejas Pastor**, C Oh, B Hong, Y Ku. "PPG wave analysis-based cardiac output estimation in off-pump coronary artery bypass surgery" *IEEE EMBC 2022*, Glasgow, United Kingdom.
8. **CA Callejas Pastor**, Y Ku, C Kwon, MW Suh. "An electronystagmogram based nystagmus detection algorithm: a machine learning approach" *2021 International Conference on Electronics, Information, and Communication (ICEIC)*, Jeju, South Korea.
9. **CA Callejas Pastor**, C Oh, B Hong, Y Ku. "Machine learning-based cardiac output estimation using non-invasive photoplethysmogram in off-pump coronary artery bypass surgery" *International Biomedical Engineering Conference 2021 (IBEC)*, Seoul, South Korea.

Patents

1. **CA Callejas Pastor**, Y Ku, M Suh. "AI-Based Vestibular Rehabilitation System Using Smartphone" In progress (2026).

2. HJ Kwon, Y Ku, **CA Callejas Pastor**. Method for setting imaging angles in cerebral angiography and computing apparatus for performing the same" Patent No. KR10-2026-0021401, South Korea (2026).
3. S Byun, M Suh, HT Ryu, JS Joo, Y Ku, **CA Callejas Pastor**, S Oh. "System and method for dizziness etiology diagnosis" Patent No. KR10-2023-0055247, South Korea (2023).
4. J Choi, Y Ku, IY Jung, **CA Callejas Pastor**. "Inclination measuring device, measuring method and computer program to measure the degree of tilt or rotation of the user's head through human body image" Patent No. KR2377364, South Korea (2020).
5. J Choi, Y Ku, IY Jung, **CA Callejas Pastor**. "Inclination measuring device, measuring method and computer program using marker" Patent No. KR2377363, South Korea (2020).